THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The LCM of $2^3 \times 3$ and $2 \times 3 \times 5$ is: 1

(a) 240 **(b)** 120 **(c)** 30 **(d)** 6

2. The LCM of 6, 8 and 12 is: 1

(a) 12 **(b)** 24 **(c)** 96 **(d)** 48

3. A quadratic polynomial with sum of zeroes $-\frac{1}{4}$ and product $\frac{1}{4}$ is: 1

(a) $4x^2 + x + 1$ **(b)** $x^2 + 4x + 1$ **(c)** $4x^2 + x - 1$ **(d)** $4x^2 - x + 1$

4. A student states: "The zeroes of $x^2 + 4$ are ± 2." The correct response is: 1

(a) Correct

(b) Incorrect — $x^2 + 4 > 0$, so it has no real zero

(c) Only −2 is a zero

(d) Only 2 is a zero

5. The value of k for which $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$ has no solution is: 1

(a) 1 **(b)** 3 **(c)** −2 **(d)** 2

6. The graph of $y = -3$ is a line: 1

(a) the x-axis **(b)** through the origin

(c) parallel to the x-axis **(d)** parallel to the y-axis

7. A student states: "$x^2 + 3x + 5 = 0$ has real roots because all its coefficients are positive." The correct response is: 1

(a) Correct **(b)** Incorrect — $D = 9 - 20 < 0$

(c) It has one root **(d)** Cannot be determined

8. Which equation has two distinct real roots? 1

(a) $2x^2 - 3x + 4 = 0$ **(b)** $x^2 + 2x + 1 = 0$ **(c)** $x^2 - 5x + 6 = 0$ **(d)** $x^2 + x + 1 = 0$

9. The 10th term of the AP $\frac{1}{2}, 1, \frac{3}{2}, \ldots$ is: 1

(a) $\frac{11}{2}$ **(b)** $\frac{9}{2}$ **(c)** 5 **(d)** 10

10. The first four terms of the AP with $a = 10$, $d = 10$ are: 1

(a) $10, 11, 12, 13$ **(b)** $10, 20, 30, 40$ **(c)** $0, 10, 20, 30$ **(d)** $10, 100, 1000, \ldots$

11. In $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{AB} = \frac{3}{5}$ and $DE = 6$ cm. Then BC is: 1

(a) 3.6 cm **(b)** 10 cm **(c)** 9 cm **(d)** 15 cm

12. Which of the following is **not** a similarity criterion? 1

(a) AA **(b)** SSS **(c)** RHS **(d)** SAS

13. $2\sin 30^\circ \cos 30^\circ$ equals: 1

(a) $\sqrt{3}$ **(b)** $\frac{\sqrt{3}}{2}$ **(c)** 1 **(d)** $\frac{1}{2}$

14. Tangents PA and PB make $\angle APB = 120^\circ$. Then $\angle AOB$ is: 1

(a) 120° **(b)** 60° **(c)** 240° **(d)** 30°

15. The arc length of a sector of radius 7 cm and angle 60° is: 1

(a) $\frac{44}{3}$ cm **(b)** $\frac{22}{3}$ cm **(c)** 11 cm **(d)** 22 cm

16. For the area of a semicircle of radius 14 cm, Student A says 308 cm²; Student B says 616 cm². Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

17. The total surface area of a solid hemisphere of radius 7 cm is: 1

(a) 308 cm² **(b)** 616 cm² **(c)** 154 cm² **(d)** 462 cm²

18. The mean of the first five even natural numbers is: 1

(a) 30 **(b)** 5 **(c)** 6 **(d)** 10

19. A student states: "The mean of grouped data found by the class-mark method is always exact." The correct response is: 1

(a) Correct
(b) Incorrect — it is approximate (frequencies assumed at class marks)
(c) Exact for odd n
(d) Cannot be determined

20. The probability of getting a 6 on a die is: 1

(a) $\frac{1}{2}$ **(b)** 1 **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{6}$

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Show that $\frac{2\sqrt{3}}{5}$ is irrational, given that $\sqrt{3}$ is irrational. 2

22. Find a quadratic polynomial whose zeroes are $2+\sqrt{3}$ and $2-\sqrt{3}$. 2

23. Solve by factorisation: $3x^2 - 2\sqrt{6}x + 2 = 0$. 2

24. Are triangles with sides 3, 4, 6 cm and 6, 8, 12 cm similar? State the criterion. 2

25. Find the point on the x-axis equidistant from $(7, 6)$ and $(3, 4)$. 2

26. If $7\sin^2\theta + 3\cos^2\theta = 4$ (θ acute), find $\tan\theta$. 2

27. Find the length of the shadow of a 6 m pole when the sun's elevation is 60°. 2

28. A tangent PQ at P of a circle of radius 9 cm meets a line through the centre at Q with $OQ = 41$ cm. Find PQ. 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $4x - y = 8$ and $2x + y = 4$. **3**

30. Find the mean of: **3**

x	5	10	15	20	25
f	4	6	10	6	4

31. How many terms of the AP $54, 51, 48, \ldots$ must be taken so that their sum is 513? Explain the double answer. **3**

32. Tangents at A and B to a circle with centre O meet at T. If $\angle ATB = 80^\circ$, find $\angle TAB$ and $\angle OAB$. **3**

33. A minute hand 21 cm long moves from 6:05 to 6:40. Find the area swept. **3**

34. Two dice are thrown. Find P(sum 12), P(sum odd) and P(product 12). **3**

35. The angles of elevation of the top of a tower from two points 30 m apart on the same side of it, in line with its foot, are 30° and 60°. Find the height of the tower. **3**

OR

From a point 20 m from the foot of a building, the angle of elevation of the top of the building is 45° and that of the top of a flagpole on it is 60°. Find the length of the flagpole. **3**

36. The mean of the following data is 50. Find p. **3**

Class	10–30	30–50	50–70	70–90
Frequency	5	8	p	7

OR

Find the mode of: **3**

Class	0–20	20–40	40–60	60–80	80–100
Frequency	10	8	12	16	4

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. A(4, 3) and B(x, 5) lie on a circle with centre O(2, 3). **5**

(i) Find x.

(ii) Find the other end of the diameter through A.

(iii) Check whether $(2, 1)$ lies on the circle.

OR

Case study: On a cricket field, the stumps are at O(0, 0) and two fielders at F_1(30, 40) and F_2(−48, 14) (in m). **5**

(i) Show that both fielders are on a circle of radius 50 m around O.

(ii) Find the distance between them.

(iii) A third fielder stands midway between them. Find his position.

38. Prove that $\frac{\tan\theta}{\sec\theta - 1} + \frac{\tan\theta}{\sec\theta + 1} = 2\csc\theta$. **5**

OR

Prove that $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\csc A - 1}{\csc A + 1}$. **5**

39. A sphere of radius 6 cm fits exactly inside a cylinder (touching its curved surface and both ends). Find **5**

(i) the volume of empty space

(ii) the ratio of the volumes of the cylinder and the sphere.

OR

A building is a cylinder of radius 3.5 m and height 6 m with a hemispherical dome. **5**

(i) Find the volume of air inside.

(ii) Find the inner surface area of walls and dome.

(iii) Find the cost of painting it at ₹10 per m^2.

40. The mean pocket allowance is ₹18. Find the missing frequency f, and state the modal class. **5**

Allowance (₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Children	7	6	9	13	f	5	4

OR

Marks of two sections are given. **5**

(i) Find the mean of each section.

(ii) Find the median of each.

(iii) Which section did better on the median?

Marks	0–10	10–20	20–30	30–40
Section A	3	7	10	5
Section B	5	5	8	7

END OF QUESTION PAPER **Total: 80 Marks**

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